

Non-equilibrium Electronic Transitions in Bulk Metals and their Contribution to Photoluminescence

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Abstract

In conductors, free electrons play a central role in photoluminescence (PL). Microscopic calculation of electronic transitions within the conduction band (intraband transitions) at thermal equilibrium successfully reproduces the famous Planckian emission. While transitions between the conduction and valence band (interband transitions) contribute negligibly to thermal emission, their role becomes significant when the system is perturbed by an external source. In this work, we seek to quantify the PL contribution of interband transitions in gold under such non-equilibrium scenarios. Utilizing previously derived non-equilibrium electron distributions—covering both steady-state continuous-wave (CW) illumination and transient regimes arising from pulsed excitation—we extend the theoretical framework to include interband channels. We validate our approach by reproducing established intraband results and systematically assessing the validity of common approximations, ultimately providing a more complete spectral description of photoluminescence in excited metals.

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1 Introduction

The emission of EM radiation from a body can be (in which cases?) attributed to photonic and electronic contributions separately [Novotny], namely

$$I(\hbar\omega) = I_{\text{ph}}(\hbar\omega) \cdot I_{\text{e}}(\hbar\omega) \quad (1.1)$$

In this study we focus on the latter. Electronic transitions which begin at a high energy state and end up in a lower energy state result in the spontaneous emission of photons with **appropriate energy**. Transitions such as these can occur either inside the conduction band (assisted by phonons) or between bands - intraband and interband transitions respectively¹.

$$I_{\text{e}} = I_{\text{cc}} + I_{\text{cv}} \quad (1.2)$$

Our materials of interest are metals (specifically gold) in which the predominant electronic contribution to emission comes from intraband transitions². Therefore, in a state of thermal equilibrium we also don't expect measurable interband presence in emission spectrum from metals [Marini?].

Note: Main Thesis

In non-equilibrium scenarios, however, such as in the steady-state case of a metal illuminated by monochromatic radiation, incoming photons of sufficient energy might be able to generate a significant hole population in the valence band which would allow for interband transitions to be more prominent in the emission spectrum of the metal.

Non-equilibrium electronic distribution for the above scenario has been derived and applied to the case of intraband transitions [Sivan & Dubi]. These calculations, however, enjoyed the approximate isotropy of the conduction band which allows one to transition to the more lenient energy space.

The valence band is generally anisotropic and when performing interband calculations one has to remain in k-space, which is what we are going to do.

Once we have established that our calculations in momentum space align with those performed in energy space - both in equilibrium and non-equilibrium settings - we proceed to perform the interband scenario. In the end, hopefully, we will be able to provide a more complete emission spectrum to both cases.

1.1 Gold Nanoparticles

We focus on interband transitions from the X and L points in k-space whose approximate symmetry can be used to simplify calculations. They also account for lion-share of the interband transitions, but not all of them.

¹Technically transitions can occur inside the valence band, but these are very unlikely and their contribution at standard cases is negligible.

²TODO: add footnote text.

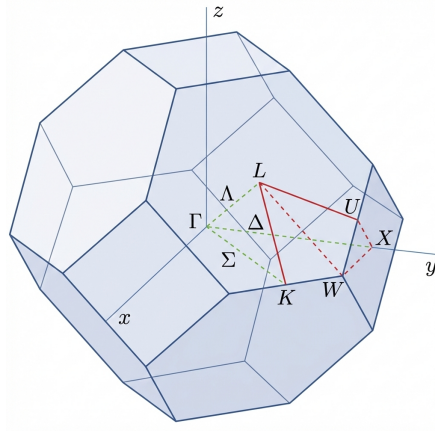


Figure 1: Pasted image 20260121090737.

The overall electronic contribution to emission is therefore given by

$$\Gamma = \Gamma_{cc} + 6\Gamma_{cv}^X + 8\Gamma_{cv}^L \quad (1.3)$$

1.2 Progress Report, Tasks & Questions

1.2.1 Progress Report

- Settled on a convergent numerical scheme for the calculation of the emission integral. ([3.1 Numeric Convergence](#))
- Established approximate nature of the analytic expressions for intraband emission (thermal and non-equilibrium). ([3.2 Analytic Approximations](#))
 - Outlined its region of validity.
 - Found an exact analytic expression for thermal intraband with constant eDOS.
- Quantified the discrepancy between constant eDOS and energy-dependent eDOS. ([Figure 9](#))
- Recognized the difficulty and inefficiency of working in k-space and having to approximate the delta function numerically ([A5 Numeric Delta Approximation](#))
- Found the proper transformations that take us to energy space, even with non-isotropic dispersion relations ([A7 Quadratic Intraband Transitions](#), [A8 Interband Transitions](#))

1.2.2 Tasks

- ☒ Improve abstract
- ☒ add a more quantitative view of the constant eDOS mismatch
- ☐ Focus on lower emission energies.
- ☐ Reproduce Rossei's absorption calculation
- ☐ Calculate intraband emission using Rossei's bands and compare the free electron approximation (parabolic band)

- ☐ Calculate interband emission using Rossei's bands and compare to absorption - in thermal eq
emission = absorption plank (Kirchhoff).
- ☐ ?

1.2.3 Questions

- Absorption and emission happen only at skin-depth. The statistical transition from a single emitter to continuum includes counting emitters in the particle's entire volume. Isn't it more appropriate to calculate specific emission?
- We're either calculating electronic transitions in bulk metals in which case we have to account for phonon-electron interaction for momentum conservation to hold.
- Or we're calculating electronic transitions in nano-particles in which a first order transition is opened due to the confinement of the wave-function in space and the spreading of momentum. (Landau damping)
- In testing theoretical predictions against experimental evidence, shouldn't the photonic DOS
- In experimental settings I imagine LDOPS (which are generally not that of vacuum?) affect the emission spectrum both qualitatively and quantitatively - how do we address this when drawing comparisons?

2 Theoretical Framework

The emission of electromagnetic radiation from a material body is governed by the intricate coupling between the body's optical environment and its intrinsic material properties. We are using Fermi's golden rule to calculate transition rates (and consequently the emission³) enables us to decouple the photonic aspect of emission from the electronic part roughly as follows

$$\Gamma(\hbar\omega) = \Gamma_{\text{ph}}(\hbar\omega) \cdot \Gamma_e(\hbar\omega) \quad (2.1)$$

The photonic part is characterized by the Local Density of States (LDOS), which encapsulates the geometry and antenna properties of the structure. The electronic contribution, which is the focus of this study, describes the material's intrinsic response.

The electronic transitions of interest are the intraband transitions inside the conduction band Γ_{cc} and the interband transition between the valence and the conduction band Γ_{cv} . Their sum represent their shared contribution to the electronic aspect of emission

$$\Gamma_e(\hbar\omega) = \Gamma_e^{\text{cc}}(\hbar\omega) + \Gamma_e^{\text{cv}}(\hbar\omega) \quad (2.2)$$

Our material of choice for this work is gold for which an approximate dispersion relation exists for both bands [Rosei]. Those approximations hold near the X and L points in Gold's Brillouin zone which has 6 and 8 such facets respectively. So a total account of the electronic part should be

$$\Gamma_e(\hbar\omega) = 6\Gamma_e^{\text{X}}(\hbar\omega) + 8\Gamma_e^{\text{L}}(\hbar\omega) \quad (2.3)$$

We focus on the high-symmetry X and L points in the Brillouin zone, which account for the majority of the interband density of states (why is that statement true).

2.1 General Emission Integral

For the electronic aspect of emission, the most general expression describes a transition from an electronic state with momentum $\mathbf{k}_1$ in energy band n , $|\psi_{n\mathbf{k}_1}\rangle$, to a state with momentum $\mathbf{k}_2$ in energy band m is as follows (see [A1 Derivation of the General Emission Integral](#))

$$\Gamma_e(\hbar\omega) \propto \sum_{n,m} \iint_{\text{BZ}} |\langle \psi_{n\mathbf{k}_1} | \hat{\mu} | \psi_{m\mathbf{k}_2} \rangle|^2 \quad (2.4)$$

$$f(\mathcal{E}_n(\mathbf{k}_1)) \left[1 - f(\mathcal{E}_m(\mathbf{k}_2)) \right] \quad (2.5)$$

$$(2.6)$$

$$\delta[\mathcal{E}_m(\mathbf{k}_2) - \mathcal{E}_n(\mathbf{k}_1) + \hbar\omega] d^3k_1 d^3k_2 \quad (2.7)$$

This is gained from FGR integrated over all initial and final states and weighted by electronic distribution which reflect occupancy f and vacancy $\bar{f}$ in the initial and final states respectively. Electronic distribution can be either thermal (Fermi-Dirac) f^T , steady-state f^S or transient f^P for CW and pulsed illumination (more on those later).

³A more rigorous treatment (other than FGR) would be required for unusually strong field strengths since we're dealing with metals which have very short coherence times.

The transition dipole moment μ^{nm} is generally momentum (energy) dependent. For interband transitions it is a common approximation to consider the TDM as constant. In the case of intraband transitions in the conduction band this is not necessarily the case. The second order TDM required for intraband transitions in bulk metals has explicit energy dependence [provide reference]. In nano-particles however, spatial confinement results in momentum uncertainty which opens a first order channel for intraband transition. This first order term, to a good approximation (according to Yonatan), has weak energy dependence and can be considered constant. We choose to work with the latter case and henceforth the TDM is treated as constant.

$$\Gamma_e(\hbar\omega) \propto \sum_{n,m} \iint_{\text{BZ}} f_n(\mathbf{k}_1) \overline{f_m(\mathbf{k}_2)} \delta(\mathcal{E}_m(\mathbf{k}_2) - \mathcal{E}_n(\mathbf{k}_1) + \hbar\omega) d^3k_1 d^3k_2 \quad (2.8)$$

2.2 Intraband Transitions

$$\Gamma_e(\hbar\omega) \propto \sum_{n,m} \iint_{\text{BZ}} f(\mathbf{k}_1) [1 - f(\mathbf{k}_2)] \delta(\mathcal{E}(\mathbf{k}_2) - \mathcal{E}(\mathbf{k}_1) + \hbar\omega) d^3k_1 d^3k_2 \quad (2.9)$$

(under what assumption) we can move to energy space as follows

$$I(\hbar\omega) \propto \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\mathcal{E}_1) [1 - f(\mathcal{E}_2)] \rho(\mathcal{E}_1) \rho(\mathcal{E}_2) \delta(\mathcal{E}_1 - \mathcal{E}_2 - \hbar\omega) d\mathcal{E}_1 d\mathcal{E}_2 \quad (2.10)$$

where

$$\rho(\mathcal{E}) = \int_{\text{BZ}} \delta(\mathcal{E} - \mathcal{E}(\mathbf{k})) d^3k \quad (2.11)$$

3 Intraband Transitions

- We now consider intraband transitions namely $\mathcal{E}_n = \mathcal{E}_m$.
- We also assume a constant transition dipole matrix $\mu_{cc}(\mathbf{k}_1, \mathbf{k}_2) \rightarrow \mu_{cc}$.

These two allow us to write the general emission integral (omitting the photonic contribution and TDM for brevity) as follows

$$I_e(\hbar\omega) \approx \iint_{\text{BZ}} f(\mathbf{k}_1) [1 - f(\mathbf{k}_2)] \delta(\mathcal{E}(\mathbf{k}_1) - \mathcal{E}(\mathbf{k}_2) - \hbar\omega) d^3k_1 d^3k_2 \quad (3.1)$$

For now we assume the free electron approximation which means our dispersion relation is isotropic and the integral can be trivially reduced to a 1D integral in energy space

$$I_e(\hbar\omega) = \int_0^\infty \int_0^\infty f(\mathcal{E}(k_1)) [1 - f(\mathcal{E}(k_2))] \delta(\mathcal{E}(k_1) - \mathcal{E}(k_2) - \hbar\omega) (4\pi k_1^2) (4\pi k_2^2) dk_1 dk_2 \quad (3.2)$$

$$= \int_0^\infty \int_0^\infty f(\mathcal{E}_1) [1 - f(\mathcal{E}_2)] \rho(\mathcal{E}_1) \rho(\mathcal{E}_2) \delta(\mathcal{E}_1 - \mathcal{E}_2 - \hbar\omega) d\mathcal{E}_1 d\mathcal{E}_2 \quad (3.3)$$

$$= \int_0^\infty f(\mathcal{E} + \hbar\omega) [1 - f(\mathcal{E})] \rho(\mathcal{E} + \hbar\omega) \rho(\mathcal{E}) d\mathcal{E} \quad (3.4)$$

Another approximation we now make is the constant electronic density of states

$$I_e(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \int_0^\infty f(\mathcal{E} + \hbar\omega) [1 - f(\mathcal{E})] d\mathcal{E} \quad (3.5)$$

For thermal electronic distribution f^T this allows us to write the emission integral in both an exact and an approximate analytic forms

$$I_e^T(\hbar\omega) = \rho^2(\mathcal{E}_F) \frac{k_B T}{e^{\beta\hbar\omega} - 1} \left[\beta\hbar\omega + \ln(1 + e^{-\beta\mu}) - \ln(1 + e^{\beta(\hbar\omega - \mu)}) \right] \quad (3.6)$$

$$\approx \rho^2(\mathcal{E}_F) \underbrace{\frac{\hbar\omega}{e^{\beta\hbar\omega} - 1}}_{\mathcal{E}_{BB}(\omega)} \quad (3.7)$$

And for a steady-state distribution f^S we can approximate the result by

$$I_e^S(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \left[\mathcal{E}_{BB}(\omega) + \delta_E A(\omega, \omega_L) + \delta_E^2 B(\omega, \omega_L) \right] \quad (3.8)$$

Where

$$A(\omega, \omega_L) = 2\mathcal{E}_{BB}(\omega - \omega_L) \quad (3.9)$$

$$B(\omega, \omega_L) = \mathcal{E}_{BB}(\omega - 2\omega_L) - 4\mathcal{E}_{BB}(\omega - \omega_L) \quad (3.10)$$

3.1 Numeric Convergence

We begin by establishing an integration scheme. For this we need to compare two identical expressions

$$\boxed{\begin{aligned} & \rho^2(\mathcal{E}_F) \int_0^\infty f^T(\mathcal{E} + \hbar\omega) [1 - f^T(\mathcal{E})] d\mathcal{E} \\ & \rho^2(\mathcal{E}_F) \frac{k_B T}{e^{\beta\hbar\omega} - 1} \left[\beta\hbar\omega + \ln(1 + e^{-\beta\mu}) - \ln(1 + e^{\beta(\hbar\omega - \mu)}) \right] \end{aligned}} \quad (3.11)$$

This serves as a good testing ground for numerical convergence as we expect to be able to reach near machine accuracy. We are going to be using relative error as our comparison metric throughout the following chapter

3.1.1 Integration Scheme

Unsurprisingly, for the same integration interval and step size, Simpson integration is much more reliable. I have also tested a quadrature which resulted in a very unstable result.

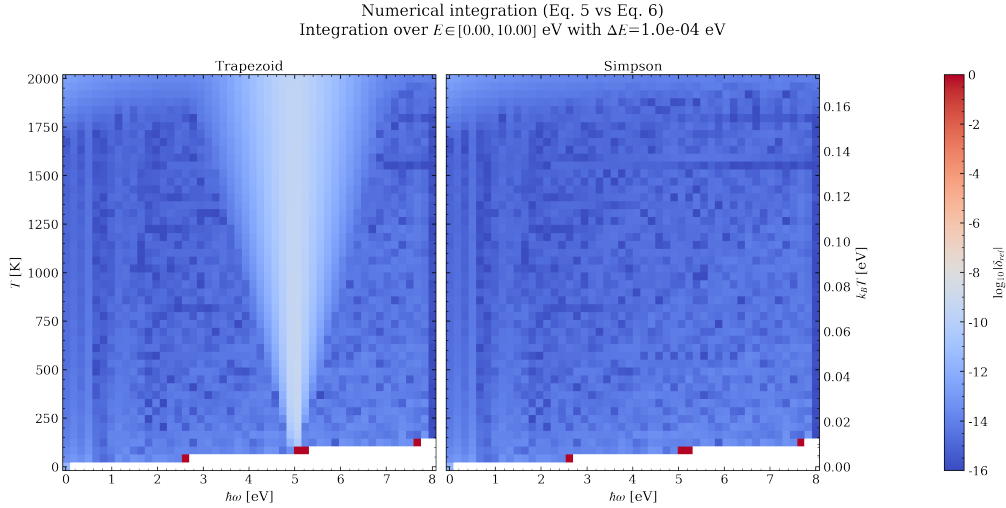


Figure 2: Figure 3.?: Point-wise relative error evaluated on the default energy-temperature grid

Figure 3.? maps the relative error across the $(T, \hbar\omega)$ plane. A distinct "cone" of higher relative error is observed expanding from the Fermi energy with increasing temperature. This behavior is expected because at large photon energies ($\hbar\omega \gg k_B T$), the transition rates are exponentially suppressed. In these regions, the relative error calculation involves dividing by a near-zero reference value, making it highly sensitive to machine epsilon and finite truncation effects. However, in the physically significant spectral region (high signal), the scheme maintains excellent accuracy.

3.1.2 Step size convergence

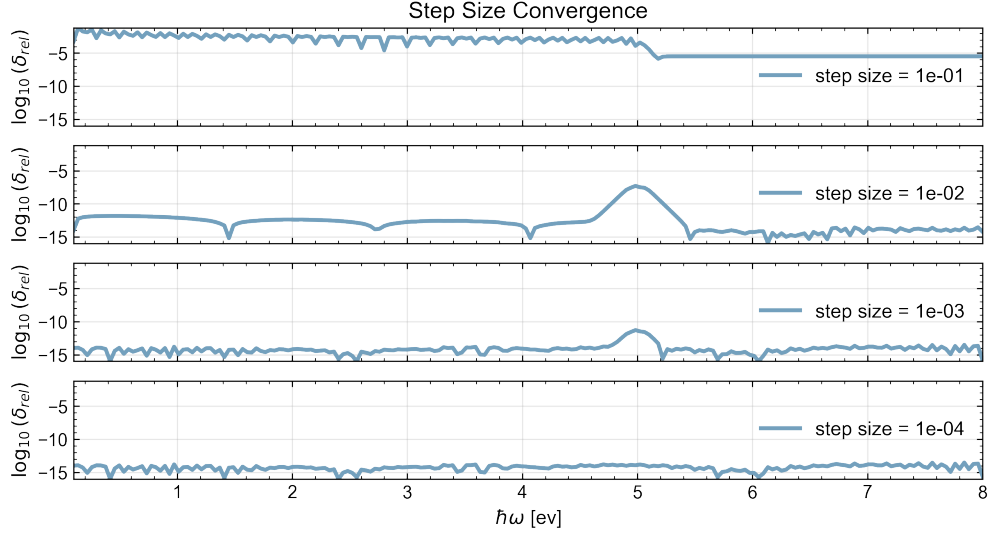


Figure 3: step_size_convergence.

The step convergence plot demonstrates that refining the grid size $\Delta\mathcal{E}$ leads to a rapid exponential reduction in error. A step size of $\Delta\mathcal{E} \approx 10^{-4}$ eV is sufficient to reach the noise floor, where the error becomes dominated by floating-point precision rather than discretization.

3.1.3 Interval Length Convergence

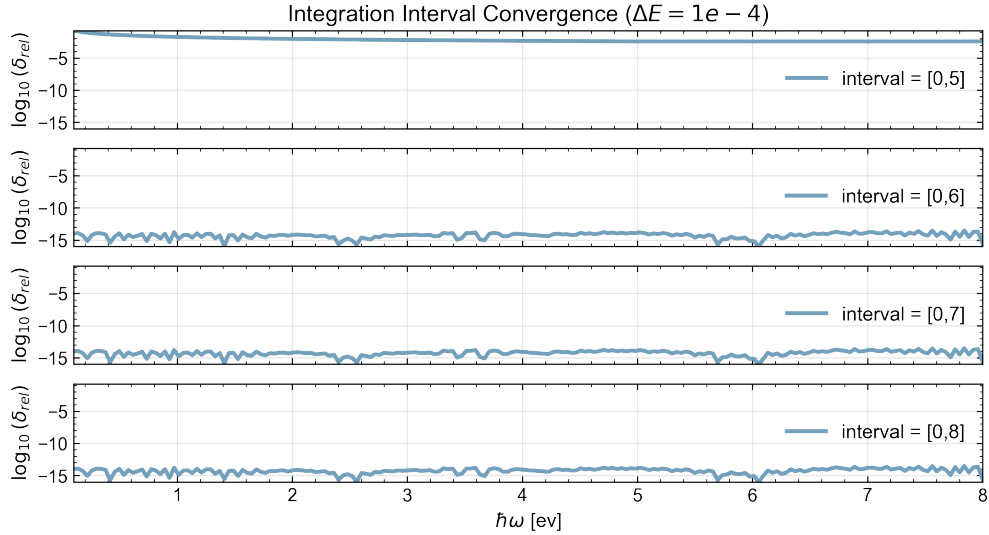


Figure 4: interval_length_convergence.

The length convergence analysis indicates that the integral value saturates quickly as the upper integration limit is increased. An upper limit of approximately 5.8 eV relative to the Fermi level is

found to be sufficient; extending the range further yields diminishing returns due to the exponential decay of the thermal occupation factors.

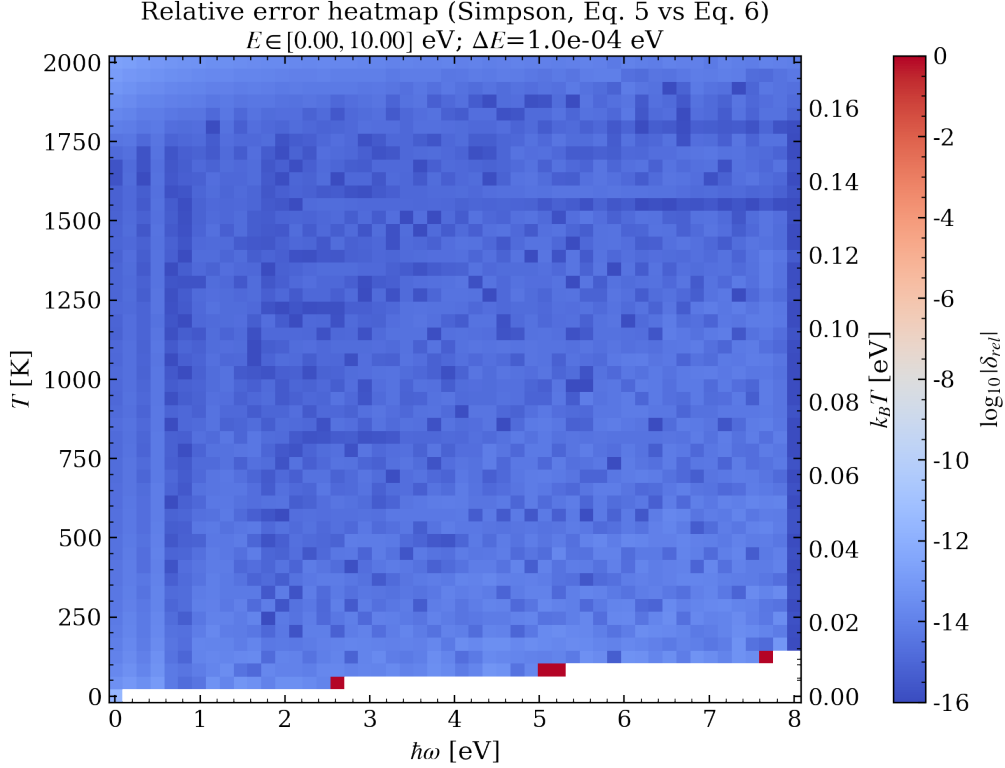


Figure 5: stage_2_rel_error_grid_50x50.

With a numerically stable and demonstrably converged treatment of Eq. (3.11) in hand, we have established robust integration parameters ($d\mathcal{E} = 10^{-4}$ eV, range limit ≈ 5.8 eV), as well as written the integrand in a more numerically stable form. We now turn to assess the validity of the physical approximations which will be the focus of the following sections.

3.1.4 Log-stable thermal factor

The factor $f^T(\mathcal{E} + \hbar\omega)[1 - f^T(\mathcal{E})]$ is evaluated in log form to avoid overflow/underflow. Let

$$a = \beta(\mathcal{E} - \mu), \quad b = a + \beta\hbar\omega. \quad (3.12)$$

Then

$$f^T(\mathcal{E} + \hbar\omega)[1 - f^T(\mathcal{E})] = \frac{e^a}{(1 + e^a)(1 + e^b)}, \quad (3.13)$$

and therefore

$$\log\left(f^T(\mathcal{E} + \hbar\omega)[1 - f^T(\mathcal{E})]\right) = a - \log(1 + e^a) - \log(1 + e^b). \quad (3.14)$$

In code, $\log(1 + e^x)$ is computed as $\text{logaddexp}(0, x)$, and the exponentiation is performed once at the end.

3.2 Analytic Approximations

After having established proper numeric convergence and extracted proper integration parameters we now turn to evaluate the region of validity of the analytic approximations.

$$I_e^T(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \int_0^\infty f^T(\mathcal{E} + \hbar\omega) [1 - f^T(\mathcal{E})] d\mathcal{E} \quad (3.15)$$

$$I^T(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \quad (3.16)$$

The relative error of Eqs. (3.15) and (3.16) above is calculated for the thermal case and for Eqs. (3.17) and (3.18) below for the steady-state case.

$$I_e^S(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \int_0^\infty f^S(\mathcal{E} + \hbar\omega) [1 - f^S(\mathcal{E})] d\mathcal{E} \quad (3.17)$$

$$I_e^S(\hbar\omega) \approx \rho^2(\mathcal{E}_F) \left[\mathcal{E}_{BB}(\omega) + \delta_E A(\omega, \omega_L) + \delta_E^2 B(\omega, \omega_L) \right] \quad (3.18)$$

In the figure below the relative error of the two cases is plotted with respect to emission energy and for several electronic temperatures each

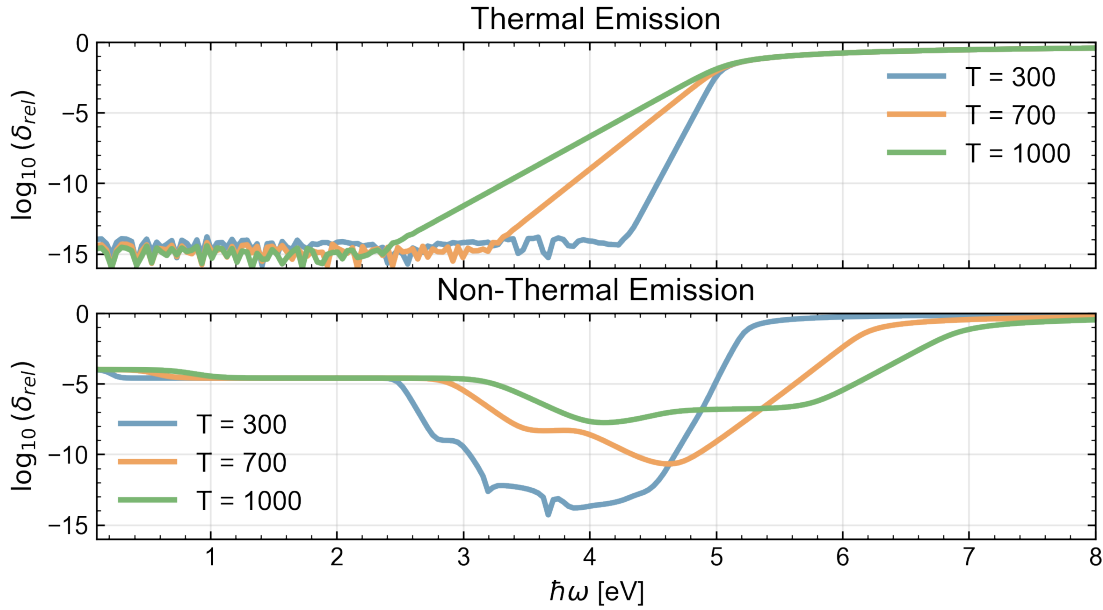


Figure 6: figure 3.2.1 Analytic approximation vs constant eDOS numeric comparison. Thermal comparison in the upper graph and non-equilibrium in the lower. Each relative error spectrum is calculated for three different temperatures.

A broader perspective can be gained by looking not only on the relative error with respect to the emission spectrum, but also with respect to electronic temperatures as can be viewed in the heatmaps below for both thermal and steady state

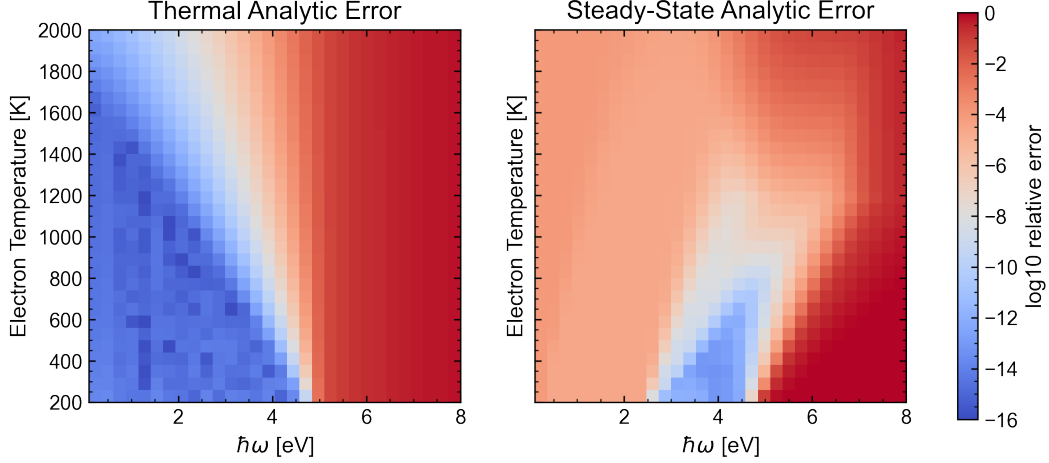


Figure 7: Figure 3.2.1: Heat maps of $\log_{10} |\delta_{\text{rel}}|$ with respect to emission energy and electron temperature for the analytic approximations of both the thermal and steady-state cases

$$\ln \left(1 + e^{\beta(\hbar\omega - \mu)} \right) \approx 0 \quad (3.19)$$

$$\iff \beta(\hbar\omega - \mu) \ll -M \quad M \in \mathbb{R} > 0 \quad (3.20)$$

$$\iff k_B T \ll \mu - \frac{1}{M} \hbar\omega \quad (3.21)$$

3.2.1 Results and discussion

If one is concerned with low electron temperatures and emission energies $\hbar\omega < \mathcal{E}_F$ the approximation. Still, it is not clear why on

- In the beginning I was calculating δ_{rel} with Eq. (3.7) and Eq. (3.8) explicitly. This created an issue for very small values of $\hbar\omega$ due to an unstable factor of $n_B(\omega)/n_B(\omega)$. This was immediately solved by simply cancelling the two terms.

In the remainder of the pipeline we will use Eq. (3.6) as the exact constant-eDOS reference. The next stage therefore establishes that the numerical evaluation of Eq. (3.5) converges to Eq. (3.6) for a practical (and not overly conservative) choice of integration step $\Delta\mathcal{E}$.

3.3 Constant eDOS Approximation

Previous section established that Eq. (3.5) can be evaluated numerically to high accuracy on a practical $\mathcal{E}$ grid. We now address the dominant approximation in the analytic pipeline the constant eDOS approximation (CEA), namely

$$\rho(\mathcal{E} + \hbar\omega)\rho(\mathcal{E}) \approx \rho^2(\mathcal{E}_F), \quad (3.22)$$

which turns Eq. (3.4) into Eq. (3.5). The equations that we're going to compare here are the two that follow

$$\boxed{\begin{aligned} & \int_0^\infty f^{T/S}(\mathcal{E} + \hbar\omega) [1 - f^{T/S}(\mathcal{E})] \rho(\mathcal{E} + \hbar\omega) \rho(\mathcal{E}) d\mathcal{E} \\ & \int_0^\infty f^{T/S}(\mathcal{E} + \hbar\omega) [1 - f^{T/S}(\mathcal{E})] \rho^2(\mathcal{E}_F) d\mathcal{E} \end{aligned}} \quad (3.23)$$

The relative error will be calculated using the usual

$$\delta_{\text{rel}}^{(3)}(\hbar\omega) = \left| \frac{I_{(5)}(\hbar\omega) - I_{(4)}(\hbar\omega)}{I_{(4)}(\hbar\omega)} \right| \quad (3.24)$$

where $I_{(4)}$ and $I_{(5)}$ denote the $\mathcal{E}$ integrals in Eq. (3.4) and Eq. (3.5), evaluated on the same numerical grid, and the T/S represent thermal and steady-state cases.

3.3.1 Direct Relative Error of the Constant eDOS Approximation

To separate the eDOS effect from the *thermal* weighting, we also examine the point-wise mismatch

$$\delta_{\text{DOS}}(\mathcal{E}, \hbar\omega) = \left| \frac{\rho(\mathcal{E} + \hbar\omega)\rho(\mathcal{E}) - \rho^2(\mathcal{E}_F)}{\rho(\mathcal{E} + \hbar\omega)\rho(\mathcal{E})} \right| = \left| 1 - \frac{\rho^2(\mathcal{E}_F)}{\rho(\mathcal{E} + \hbar\omega)\rho(\mathcal{E})} \right|. \quad (3.25)$$

For the free-electron DOS $\rho(\mathcal{E}) \propto \sqrt{\mathcal{E}}$ this reduces to

$$\delta_{\text{DOS}}(\mathcal{E}, \hbar\omega) = \left| 1 - \frac{\mathcal{E}_F}{\sqrt{\mathcal{E}(\mathcal{E} + \hbar\omega)}} \right|, \quad (3.26)$$

so the locus of minimal DOS mismatch satisfies

$$\mathcal{E}(\mathcal{E} + \hbar\omega) = \mathcal{E}_F^2. \quad (3.27)$$

Which can be seen in the figure below

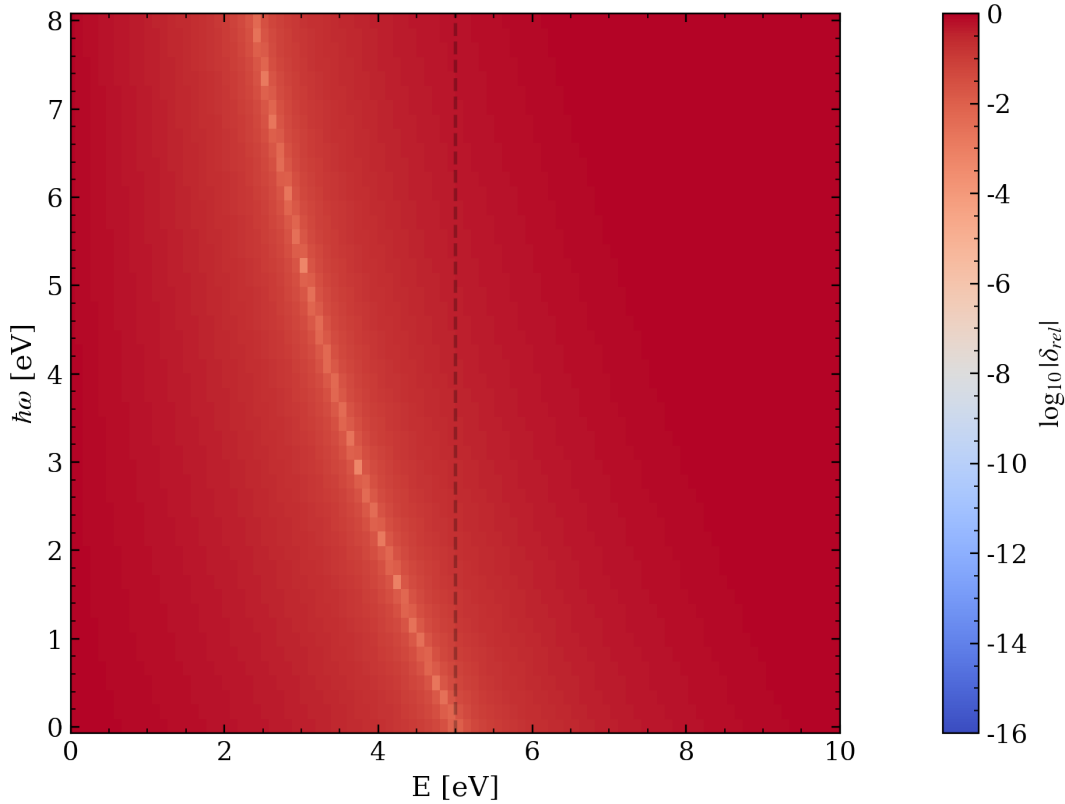


Figure 8: Figure 3.1: Heat map of $\log_{10} \delta_{\text{DOS}}(\mathcal{E}, \hbar\omega)$ for $\rho(\mathcal{E})\rho(\mathcal{E} + \hbar\omega)$ relative to $\rho^2(\mathcal{E}_F)$, evaluated for $\mathcal{E}_F = 5$ eV.

Figure 3.1 shows that the constant-DOS-product assumption is only plausible in regions where the geometric mean $\sqrt{\mathcal{E}(\mathcal{E} + \hbar\omega)}$ remains close to $\mathcal{E}_F$. For fixed $\mathcal{E}_F$, large $\hbar\omega$ forces one of the two energies to be far from the other, and for small $\mathcal{E}$ the free-electron DOS varies rapidly and the mismatch necessarily grows.

3.3.2 Emission Relative Error Due to Constant eDOS Approximation

The physically relevant quantity, however, is the $\mathcal{E}$ -integrated error (Figures 3.2 and 3.3), where $\delta_{\text{DOS}}(\mathcal{E}, \hbar\omega)$ is weighted by the thermal factor $f^T(\mathcal{E} + \hbar\omega)[1 - f^T(\mathcal{E})]$. Since this thermal factor localizes the integral to an energy window of width $\mathcal{O}(k_B T)$, the approximation quality depends on whether that window samples energies for which $\rho(\mathcal{E})$ is approximately constant on the relevant scale.

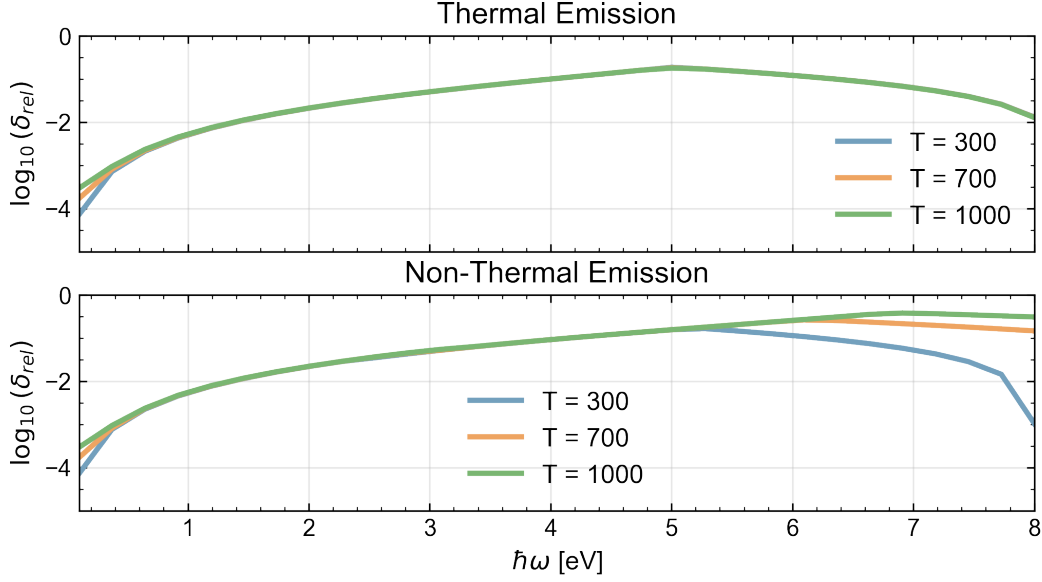


Figure 9: Figure 3.2 - Relative error in log-scale over the emission spectrum for several temperatures (thermal on the top steady-state in the bottom)

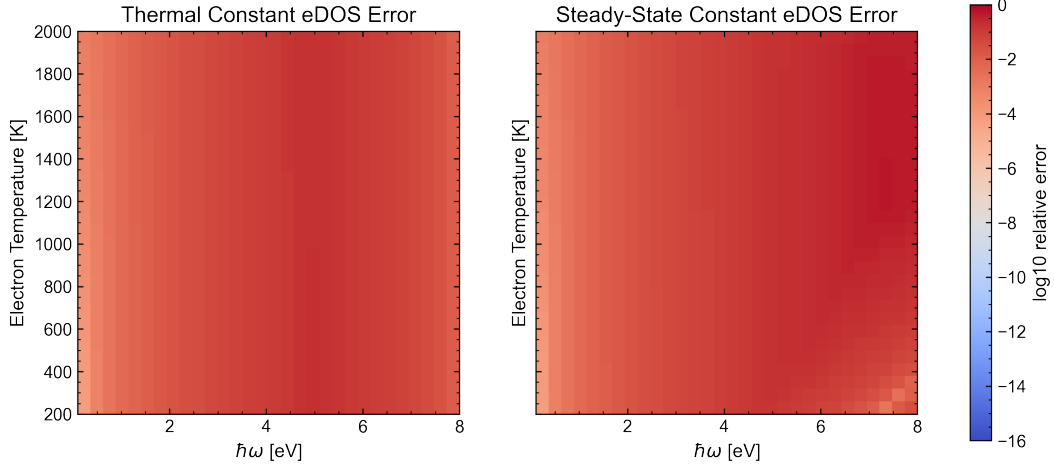


Figure 10: Figure 3.3 - Relative error in log-scale wrt the emission spectrum and temperature for both thermal and steady-state cases for a more comprehensive comparison

It appears as though the constant eDOS approximation is not very accurate. Need to discuss with Yonatan whether this is something he accepts, something I missed or perhaps something that should be fixed.

Having quantified the constant-eDOS approximation, the remaining stages address the earlier steps in the derivation—most notably the delta-function resolution that connects Eq. (3.3) to Eq. (3.4).

3.4 Delta Approximation

3.4.1 Gaussian Delta Approximation (Double Integral)

We approximate the energy conservation delta function $\delta(E_1 - E_2 - \hbar\omega)$ with a Gaussian $G_\sigma(x)$ of width σ . To optimize performance, the integration over the energy difference $u = E_1 - E_2 - \hbar\omega$ is restricted to a window $[-M\sigma, M\sigma]$ with N steps within 2σ :

$$I(\hbar\omega) \approx \int \rho(\mathcal{E}_1) f(\mathcal{E}_1) \left[\int_{-M\sigma}^{M\sigma} \rho(\mathcal{E}') [1 - f(\mathcal{E}')] G_\sigma(u) du \right] d\mathcal{E}_1 \quad (3.28)$$

where $\mathcal{E}' = \mathcal{E}_1 - \hbar\omega - u$. The parameters M and N allow tuning the balance between accuracy and computational cost.

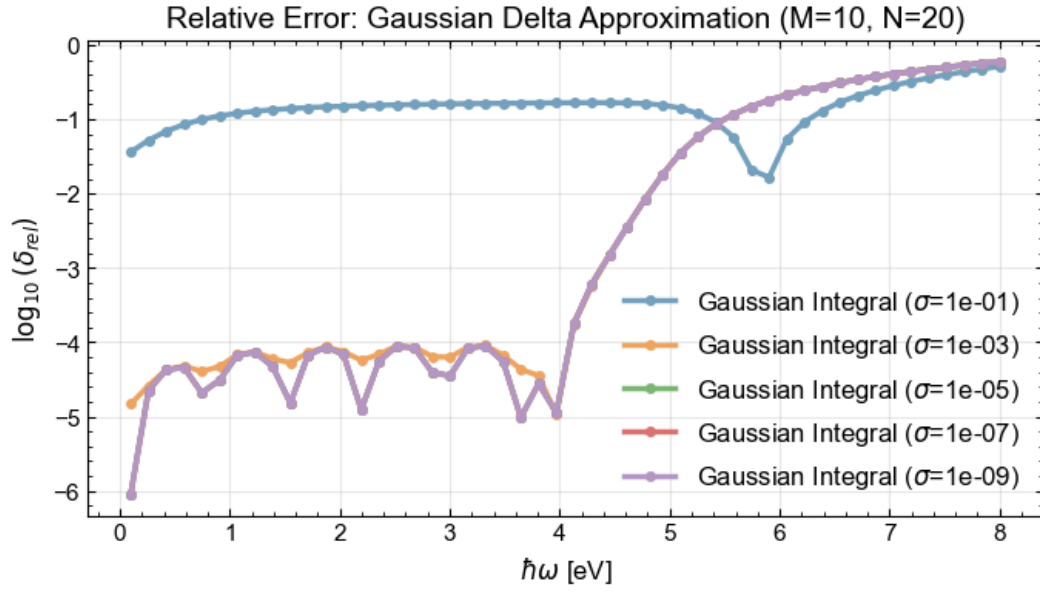


Figure 11: gaussian_delta_double_integral.

3.5 Quadratic Band

4 Interband Transitions

A1 Derivation of the General Emission Integral

A1.1 First Order Radiative Transitions in Solids

Consider electrons in a solid. It's Hamiltonian consists of the kinetic energy term and the lattice term which embodies the periodic nature of the material. In the presence of an E&M field the Hamiltonian is

$$\hat{H} = \frac{[\hat{\mathbf{p}} + e\mathbf{A}(\hat{\mathbf{r}}, t)]^2}{2m_e} + V_{\text{lattice}}(\hat{\mathbf{r}}) \quad (\text{A1.1})$$

$$\approx \underbrace{\frac{\hat{\mathbf{p}} \cdot \hat{\mathbf{p}}}{2m_e}}_{\hat{H}_0} + V_{\text{lattice}}(\hat{\mathbf{r}}) + \underbrace{\frac{e}{m_e} [\mathbf{A}(\hat{\mathbf{r}}, t) \cdot \hat{\mathbf{p}} + \hat{\mathbf{p}} \cdot \mathbf{A}(\hat{\mathbf{r}}, t)]}_{\hat{H}_{\text{int}}} \quad (\text{A1.2})$$

The diamagnetic terms, $|\mathbf{A}(\hat{\mathbf{r}}, t)|^2$, term relates to a two-photon process which is negligible in the case of spontaneous emission and we take the common course of neglecting it.

- **nonlinear effects for strong laser pumps pulsed dynamics? Well, for pulsed radiation $\mathbf{A}(\hat{\mathbf{r}}, t)$ would be different).**
- Coulomb gauge - always have this freedom?
- Need to also account for phonon part in Hamiltonian for intraband transitions?

Under the Coulomb gauge $\nabla \cdot \mathbf{A}(\hat{\mathbf{r}}, t) = 0$ the interaction term can be written as follows

$$\hat{H}_{\text{int}} = \frac{e}{m_e} \mathbf{A}(\hat{\mathbf{r}}, t) \cdot \hat{\mathbf{p}} \quad (\text{A1.3})$$

$$= \frac{eA_0}{2m_e} \left[e^{i(\mathbf{q} \cdot \mathbf{r} - \omega t)} + e^{-i(\mathbf{q} \cdot \mathbf{r} - \omega t)} \right] \hat{\mathbf{n}} \cdot \hat{\mathbf{p}} \quad (\text{A1.4})$$

$$\equiv \hat{V}_{\uparrow} e^{-i\omega t} + \hat{V}_{\downarrow} e^{i\omega t} \quad (\text{A1.5})$$

where the vector potential is that of a plane wave

$$\mathbf{A}(\hat{\mathbf{r}}, t) = \hat{\mathbf{n}} A_0 \cos(\mathbf{q} \cdot \mathbf{r} - \omega t) \quad (\text{A1.6})$$

The eigenstates of $\hat{H}_0$ are famously Bloch states $|\psi_{n\mathbf{k}}\rangle$. We treat the interaction term as a perturbation and, using Fermi Golden Rule (FGR), calculate the transition rates between energy states, namely

$$\Gamma_{\uparrow/\downarrow} = \frac{2\pi}{\hbar} \left| \left\langle \psi_{m\mathbf{k}_2} \left| \hat{V}_{\uparrow/\downarrow} \right| \psi_{n\mathbf{k}_1} \right\rangle \right|^2 \delta(\mathcal{E}_m(\mathbf{k}_2) - (\mathcal{E}_n(\mathbf{k}_1) \pm \hbar\omega)) \quad (\text{A1.7})$$

$$= \frac{2\pi}{\hbar} |\mathbf{P}_{mn} \cdot \hat{\mathbf{n}}|^2 \delta(\mathbf{k}_2 - (\mathbf{k}_1 \pm \mathbf{q})) \delta(\mathcal{E}_m(\mathbf{k}_2) - (\mathcal{E}_n(\mathbf{k}_1) \pm \hbar\omega)) \quad (\text{A1.8})$$

Where $\uparrow$ stands for absorption and $\downarrow$ for emission⁴.

⁴But shouldn't emission here be mostly spontaneous emission? The way it is presented here, stimulated absorption and emission are the opposites. It is really stimulated emission? Or are we actually calculating spontaneous emission without understanding it?

A1.2 Second Order Radiative Transitions in Solids

Note: Transition Dipole Matrix (TDM)

$$\mu_{mn} = \langle \psi_{m\mathbf{k}_2} | \hat{V}_{\uparrow/\downarrow} | \psi_{n\mathbf{k}_1} \rangle = | \langle \psi_{m\mathbf{k}_2} | e^{\pm i\mathbf{q}\cdot\mathbf{r}} \mathbf{p} \cdot \hat{\mathbf{n}} | \psi_{n\mathbf{k}_1} \rangle | \quad (\text{A1.9})$$

Bloch states are explicitly

$$| \psi_{n\mathbf{k}} \rangle = u_{n\mathbf{k}}(\mathbf{r}) \cdot e^{i\mathbf{k}\cdot\mathbf{r}} \quad (\text{A1.10})$$

where $u_{n\mathbf{k}}(\mathbf{r})$ has the periodicity of the lattice. The TDM is then explicitly

$$\int_{\text{BZ}} d^3r \left(u_{m\mathbf{k}_2}^*(\mathbf{r}) \cdot e^{-i\mathbf{k}_2\cdot\mathbf{r}} \right) (e^{\pm i\mathbf{q}\cdot\mathbf{r}} \mathbf{p} \cdot \hat{\mathbf{n}}) \left(u_{n\mathbf{k}_1}(\mathbf{r}) \cdot e^{i\mathbf{k}_1\cdot\mathbf{r}} \right) \quad (\text{A1.11})$$

$$\int d^3r e^{i(\mathbf{k}_1 - \mathbf{k}_2 \pm \mathbf{q})\cdot\mathbf{r}} u_{m\mathbf{k}_2}^*(\mathbf{r}) u_{n\mathbf{k}_1}(\mathbf{r}) \mathbf{p} \cdot \hat{\mathbf{n}} \quad (\text{A1.12})$$

Photons carry minuscule momentum ($\mathbf{q} \approx 0$) so it is very common and a good approximation to assume that $\mathbf{k}_1 \approx \mathbf{k}_2$ to simplify calculation while preserving meaningful and qualitatively accurate results. The transition rate then simplifies to the following

$$\Gamma_{\uparrow/\downarrow} = \frac{2\pi}{\hbar} \mathbf{P}_{mn} \cdot \hat{\mathbf{n}} \delta(\mathcal{E}_m(\mathbf{k}) - (\mathcal{E}_n(\mathbf{k}) \pm \hbar\omega)) \quad (\text{A1.13})$$

Looking at this equation, it is clear that such a single-process transition is possible only when $n \neq m$, namely when the energy states (bands) are different - an interband transition. It has been known for several decades that absorption and emission of photons by conduction electrons is impossible via a single-step process. It is possible however, via a two-step process in which the interaction is mediated by a phonon which provides the missing momentum.

A1.3 From a Single Transition to the Continuum

Consider an electronic transition between an initial state $|\mathbf{k}_1, \alpha\rangle$ and a final state $|\mathbf{k}_2, \beta\rangle$, mediated by an energy transfer $\hbar\omega$. The subscripts α and β represent specific energy bands (valence or conduction), allowing transitions between different branches of the dispersion relation. According to Fermi's Golden Rule, the microscopic transition rate is

$$\Gamma_{1 \rightarrow 2}(\hbar\omega) = \frac{2\pi}{\hbar} \left| \mu_{12}^{\alpha\beta} \right|^2 \delta(\mathcal{E}_\beta(\mathbf{k}_2) - \mathcal{E}_\alpha(\mathbf{k}_1) + \hbar\omega) \quad (\text{A1.14})$$

where $\mu_{12}^{\alpha\beta}$ is the transition dipole moment governing momentum conservation, and the Dirac delta function enforces energy conservation based on the band-specific dispersion relations $\mathcal{E}_\alpha$ and $\mathcal{E}_\beta$.

For clarity, we temporarily suppress the band indices to derive the rate for a generic pair of bands. To determine the rate Γ_1 from a specific initial state $|\mathbf{k}_1\rangle$, we sum over all available final states $|\mathbf{k}_2\rangle$. This summation is weighted by the probability of the final state being vacant, $[1 - f(\mathbf{k}_2)]$, and accounts for spin degeneracy. In the continuum limit, the discrete summation over wave-vectors transforms into an integral over the Brillouin Zone (BZ) with a density of states factor $V/(2\pi)^3$:

$$\Gamma_1(\hbar\omega) = 2 \cdot \sum_{\mathbf{k}_2} \Gamma_{1 \rightarrow 2}(\hbar\omega) [1 - f(\mathbf{k}_2)] \quad (\text{A1.15})$$

$$= 2V \int_{\text{BZ}} \frac{d^3 k_2}{(2\pi)^3} \Gamma_{1 \rightarrow 2}(\hbar\omega) [1 - f(\mathbf{k}_2)] \quad (\text{A1.16})$$

Similarly, to obtain the total rate for this band pair, we sum the single-state rates over all possible initial states $|\mathbf{k}_1\rangle$, weighted by the occupation probability $f(\mathbf{k}_1)$

$$\Gamma(\hbar\omega) = 2 \cdot \sum_{\mathbf{k}_1} \Gamma_1(\hbar\omega) f(\mathbf{k}_1) \quad (\text{A1.17})$$

$$= 2V \int_{\text{BZ}} \frac{d^3 k_1}{(2\pi)^3} \Gamma_1(\hbar\omega) f(\mathbf{k}_1) \quad (\text{A1.18})$$

Finally, to account for all possible emission channels, we reintroduce the band indices and perform a general summation over all bands α and β . This covers all permutations, including interband (conduction-valence $\alpha\beta = cv$) and intraband (conduction-conduction $\alpha\beta = cc$) transitions. Defining $\overline{f_\beta(\mathbf{k})} \equiv 1 - f_\beta(\mathbf{k})$, the general expression is

$$\Gamma(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2V}{(2\pi)^3} \right)^2 \sum_{\alpha, \beta} \iint_{\text{BZ}} |\mu^{\alpha\beta}(\mathbf{k}_1, \mathbf{k}_2)|^2 f_\alpha(\mathbf{k}_1) \overline{f_\beta(\mathbf{k}_2)} \delta(\mathcal{E}_\beta(\mathbf{k}_2) - \mathcal{E}_\alpha(\mathbf{k}_1) + \hbar\omega) d^3 k_1 d^3 k_2$$

(A1.19)

A2 Derivation of Analytic Approximation

A2.1 Intraband Emission

$$\Gamma_{\alpha\beta}(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2V}{(2\pi)^3} \right)^2 \iint_{BZ} |\mu(\mathbf{k}_i, \mathbf{k}_f)|^2 f[\mathcal{E}_\alpha(\mathbf{k}_i)] \overline{f[\mathcal{E}_\beta(\mathbf{k}_f)]} \delta(\mathcal{E}_\alpha(\mathbf{k}_i) - \mathcal{E}_\beta(\mathbf{k}_f) - \hbar\omega) dk_i^3 dk_f^3 \quad (\text{A2.1})$$

Note: Impossibility of a one step intraband transition

The transition dipole moment $\mu(\mathbf{k}_i, \mathbf{k}_f)$ implicitly contains a statement about the conservation of momentum, namely that

$$\mathbf{k}_i = \mathbf{k}_f + \mathbf{q} = \mathbf{k}_f \quad (\text{A2.2})$$

Where $\mathbf{q}$ is the momentum of the emitted photon. Since photons have negligible momenta relative to electrons' ($\mathbf{q} \ll \mathbf{k}$), the electron momentum should virtually remain unchanged

$$\delta(\mathbf{k}_i - \mathbf{k}_f - \mathbf{q}) \rightarrow \delta(\mathbf{k}_i - \mathbf{k}_f) \quad (\text{A2.3})$$

This result renders single process intraband transitions moot. These have zero probability of occurring. In Yonatan's calculations in energy space momentum is not conserved. The only reason why this would be ok (as I understand it) is if a two-step process is implicitly assumed in which for every energetic transition there exists a phonon which provides the exact amount of the missing momentum. This raises a couple of questions:

1. Is this even physically accurate assumption to make?
2. Even if it is, would every energetic transition have a proper phonon with the same probability? The current calculation would treat it as such.

Bottom line is that, in my opinion, intraband transitions should be treated as two-step processes, and the interband calculations - under the assumption of direct transition - would become simpler.

- For intraband transition $\alpha = \beta = c$ where c stands for conduction band.
- Constant transition dipole moment approximation $|\mu(\mathbf{k}_i, \mathbf{k}_f)|^2 \rightarrow |\mu_{cc}|^2$ (for the conduction band).

$$\Gamma_{cc}(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2\mu_{cc}V}{(2\pi)^3} \right)^2 \iint_{BZ} f[\mathcal{E}_c(\mathbf{k}_i)] \overline{f[\mathcal{E}_c(\mathbf{k}_f)]} \delta(\mathcal{E}_c(\mathbf{k}_i) - \mathcal{E}_c(\mathbf{k}_f) - \hbar\omega) d^3k_i d^3k_f \quad (\text{A2.4})$$

The dispersion relation of the conduction band is isotropic. This allows us to move from the 3-dimensional momentum space to the 1-dimensional energy space using the A4 - Electronic Density of States, namely

$$d^3k \rightarrow \rho(\mathcal{E})d\mathcal{E} = \frac{4\pi\sqrt{2m_e^3\mathcal{E}}}{\hbar^3} d\mathcal{E} \quad (\text{A2.5})$$

$$\Gamma_{cc}(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2\mu_{cc}V}{(2\pi)^3} \right)^2 \iint_{BZ} f(\mathcal{E}_{ci}) \overline{f(\mathcal{E}_{cf})} \delta(\mathcal{E}_{ci} - \mathcal{E}_{cf} - \hbar\omega) \rho(\mathcal{E}_{ci})\rho(\mathcal{E}_{cf})d\mathcal{E}_{ci}d\mathcal{E}_{cf} \quad (\text{A2.6})$$

In energy space, the delta can be easily resolved

$$\Gamma_{cc}(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2\mu_{cc}V}{(2\pi)^3} \right)^2 \int_{BZ} f(\mathcal{E}_{ci}) \overline{f(\mathcal{E}_{ci} - \hbar\omega)} \rho(\mathcal{E}_{ci})\rho(\mathcal{E}_{ci} - \hbar\omega)d\mathcal{E}_{ci} \quad (\text{A2.7})$$

Equation (A2.7) serves as our reference, since it is calculated in Yonatan's paper and is assumed to be correct due to its alignment with experimental evidence. Since we're later going approximate the delta function by a Gaussian, this is the place to establish the validity of such approximation

$$\Gamma_{\text{cc}}(\hbar\omega) = \frac{2\pi}{\hbar} \left(\frac{2\mu_{\text{cc}}V}{(2\pi)^3} \right)^2 \iint_{\text{BZ}} f(\mathcal{E}_{\text{ci}}) \overline{f(\mathcal{E}_{\text{cf}})} \lim_{\sigma_{\mathcal{E}} \rightarrow 0} G(\mathcal{E}_{\text{ci}} - \mathcal{E}_{\text{cf}} - \hbar\omega ; \sigma_{\mathcal{E}}) \rho(\mathcal{E}_{\text{ci}}) \rho(\mathcal{E}_{\text{cf}}) d\mathcal{E}_{\text{ci}} d\mathcal{E}_{\text{cf}} \quad (\text{A2.8})$$

$$G(\mathcal{E}_{\text{ci}} - \mathcal{E}_{\text{cf}} - \hbar\omega ; \sigma_{\mathcal{E}}) = \quad (\text{A2.9})$$

Under the assumption that the integral converges, the limit can be taken out of the integral

A3 Thermal Factor Energy Space

A3.1 Equilibrium

$$I = \int_0^{\infty} f(\mathcal{E} + \hbar\omega)[1 - f(\mathcal{E})]d\mathcal{E} \quad (\text{A3.1})$$

$$= n_B(\hbar\omega) \left[\int_0^{\infty} f(\mathcal{E})d\mathcal{E} - \int_0^{\infty} f(\mathcal{E} + \hbar\omega)d\mathcal{E} \right] \quad (\text{A3.2})$$

$$= n_B(\hbar\omega) \left[\int_0^{\infty} f(\mathcal{E})d\mathcal{E} - \int_{\hbar\omega}^{\infty} f(\mathcal{E})d\mathcal{E} \right] \quad (\text{A3.3})$$

$$= n_B(\hbar\omega) \int_0^{\hbar\omega} f(\mathcal{E})d\mathcal{E} \quad (\text{A3.4})$$

$$= n_B(\hbar\omega) \int_0^{\hbar\omega} \frac{1}{e^{\beta(\mathcal{E}-\mu)} + 1} d\mathcal{E} \quad (\text{A3.5})$$

$$= n_B(\hbar\omega) \int_0^{\hbar\omega} \left[1 + \frac{e^{\beta(\mathcal{E}-\mu)}}{e^{\beta(\mathcal{E}-\mu)} + 1} \right] d\mathcal{E} \quad (\text{A3.6})$$

$$= n_B(\hbar\omega) \left[\mathcal{E} - \frac{1}{\beta} \ln \left(e^{\beta(\mathcal{E}-\mu)} + 1 \right) \right]_{\mathcal{E}=0}^{\mathcal{E}=\hbar\omega} \quad (\text{A3.7})$$

$$= n_B(\hbar\omega) \left[\hbar\omega + \frac{1}{\beta} \ln \left(1 + e^{-\beta\mu} \right) - \frac{1}{\beta} \ln \left(1 + e^{\beta(\hbar\omega-\mu)} \right) \right] \quad (\text{A3.8})$$

$$\approx \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \quad (\text{A3.9})$$

The transition from Eq. (A3.1) to Eq. (A3.2) is a known identity

$$f(\mathcal{E} + \hbar\omega)[1 - f(\mathcal{E})] = n_B(\hbar\omega) [f(\mathcal{E}) - f(\mathcal{E} + \hbar\omega)] \quad (\text{A3.10})$$

where $n_B(\hbar\omega)$ is the Bose-Einstein distribution

$$n_B(\hbar\omega) = \frac{1}{e^{\beta\hbar\omega} - 1} \quad (\text{A3.11})$$

The approximation from Eq. (A3.8) to Eq. (A3.9) is valid when the following conditions are met

$$\mu \gg k_B T \quad \text{and} \quad \mu \gg \hbar\omega \quad (\text{A3.12})$$

A3.2 Non-equilibrium

$$f(\mathcal{E} \mid \omega_L) = f^T(\mathcal{E}) + \delta E(\omega_L) \cdot B(\mathcal{E} \mid \omega_L) \quad (\text{A3.13})$$

$$B(\mathcal{E} \mid \omega_L) = f^T(\mathcal{E} - \hbar\omega_L) [1 - f^T(\mathcal{E})] - f^T(\mathcal{E}) [1 - f^T(\mathcal{E} + \hbar\omega_L)] \quad (\text{A3.14})$$

$$I(\hbar\omega_E|\hbar\omega_L) \propto \int_0^\infty f(\mathcal{E} + \hbar\omega)[1 - f(\mathcal{E})]d\mathcal{E} \quad (\text{A3.15})$$

A4 Electronic Density of States

The change in variables in 3D goes as follows

$$d^3k \rightarrow 4\pi k^2 dk \rightarrow \underbrace{4\pi k^2 \frac{dk}{d\mathcal{E}}}_{\rho(\mathcal{E})} d\mathcal{E} \quad (\text{A4.1})$$

the momentum energy relation comes from the dispersion relation of the electron which is approximated to be that of a free electron for a conduction electron in the 1st BZ.

$$\mathcal{E}(k) = \frac{\hbar^2 k^2}{2m} \quad \rightarrow \quad k(\mathcal{E}) = \frac{\sqrt{2m\mathcal{E}}}{\hbar} \quad \rightarrow \quad \frac{dk}{d\mathcal{E}} = \frac{1}{\hbar} \sqrt{\frac{m}{2\mathcal{E}}} \quad (\text{A4.2})$$

plugging the relevant expressions in Eq. (A4.2) into Eq. (A4.1) results in

$$\rho(\mathcal{E}) = \frac{4\pi m_e^{3/2}}{\hbar^3} \sqrt{2\mathcal{E}} \quad (\text{A4.3})$$

A4.1 A general Form of DOS

$$\rho(\mathcal{E}) = \int_{\text{BZ}} d^3k \, \delta(\mathcal{E} - \mathcal{E}(\mathbf{k})) \quad (\text{A4.4})$$

$$= \int_{\Omega} d^3k' \, \frac{\delta(\mathbf{k} - \mathbf{k}')}{|\nabla_{\mathbf{k}} \mathcal{E}|} \quad (\text{A4.5})$$

Where Ω is the surface on which $\mathcal{E}(\mathbf{k}) = \mathcal{E}$

A5 Numeric Delta Approximation

We attempt to numerically evaluate an integral with a delta function approximated by a gaussian with vanishing width. The Gaussian is given in a standard fashion

$$G_\sigma(x|\mu) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)^2} \quad (\text{A5.1})$$

The exact, approximated and numeric integral values are given as follows

$$f(\mu) = \int_{-\infty}^{\infty} f(x)\delta(x-\mu) dx \quad (\text{A5.2})$$

$$f_\sigma^{\text{approx}}(\mu) = \int_{-\infty}^{\infty} f(x) \cdot G_\sigma(x|\mu) dx \quad (\text{A5.3})$$

$$f_{\sigma,N}^{\text{numeric}}(x_0) = \sum_{n=0}^N f(x_n) \cdot G_\sigma(x_n|\mu) \cdot \Delta x \quad (\text{A5.4})$$

Where $x_n = -L + n\Delta x$ and $\Delta x = \frac{2L}{N}$. Below is the order of convergence

$$f(x_0) = \lim_{\sigma \rightarrow 0} f_\sigma^{\text{approx}}(x_0) = \lim_{\substack{\sigma \rightarrow 0 \\ N \rightarrow \infty}} f_{\sigma,N}^{\text{numeric}}(x_0) \quad (\text{A5.5})$$

- it is worth noting that the infinite integration interval introduces yet another limit, which we don't rigorously consider, just take an interval range $2L$ that is much larger than 2σ .

$$E_{N,\sigma}^{\text{total}} = f - f_{\sigma,N}^{\text{numeric}} \quad (\text{A5.6})$$

$$= \underbrace{f - I_\sigma^{\text{approx}}}_{E_\sigma^{\text{approx}}} + \underbrace{I_\sigma^{\text{approx}} - f_{\sigma,N}^{\text{numeric}}}_{E_{N,\sigma}^{\text{numeric}}} \quad (\text{A5.7})$$

The general deviation of the calculation from the desired result has two contributions, one The numeric error of a trapezoid integration is known and given by

$$E_{N,\sigma}^{\text{numeric}} = \frac{F''_\sigma(\xi)(2L)^3}{12N^2} \quad (\text{A5.8})$$

where $\xi \in (-L, L)$. We can set an upper bound for the error by seeking a certain x^* such that $F_\sigma(x^*) \geq F_\sigma(x) \quad \forall x \in (-L, L)$. I'm not sure that E_σ^{approx} has a form that can be found, but it is clear that it depends only on σ . For a given σ

$$E_{N,\sigma}^{\text{total}} = E_\sigma^{\text{approx}} + \frac{F''_\sigma(\xi)(2L)^3}{12N^2} \quad (\text{A5.9})$$

$$\delta_{N,\sigma}^{\text{total}} = \delta_\sigma^{\text{approx}} + \frac{F''_\sigma(\xi)(2L)^3}{f(x_0) 12N^2} \quad (\text{A5.10})$$

So for a given σ , as N increases, $\delta_{N,\sigma}^{\text{total}}$ shrinks as $\propto N^{-2}$ and reaches $\delta_\sigma^{\text{approx}}$ asymptotically.

On the one hand, smaller σ corresponds to smaller $\delta_\sigma^{\text{approx}}$ value. On the other hand, since F'' contains a G'' component (which scales with σ^{-4}) we expect the numeric convergence to become more gradual as σ shrinks - namely, to require larger values of N to reach numeric accuracy. This interplay between the approximation error and the numerical error requires us to simultaneously decrease σ while increasing N for the total error to negligible (?) values.

A5.1 2D delta

$$\delta \left(\frac{\hbar^2 k_1^2}{2m} - \frac{\hbar^2 k_2^2}{2m} - \hbar\omega \right) = \lim_{\sigma \rightarrow 0} \frac{1}{\sigma \mathcal{E} \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{\frac{\hbar^2 k_1^2}{2m} - \frac{\hbar^2 k_2^2}{2m} - \hbar\omega}{\sigma \mathcal{E}} \right)^2 \right] \quad (\text{A5.11})$$

$$= \lim_{\sigma \rightarrow 0} \frac{1}{\sigma \mathcal{E} \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{k_1^2 - k_2^2 - \frac{2m\omega}{\hbar}}{\sigma_k^2} \right)^2 \right] \equiv \lim_{\sigma \rightarrow 0} G_\sigma(k_1, k_2; \omega) \quad (\text{A5.12})$$

$$I = \iint \underbrace{f(k_1, k_2) G_\sigma(k_1, k_2)}_{\substack{\text{includes Jacobian} \\ F_\sigma(k_1, k_2)}} dk_1 dk_2 \quad (\text{A5.13})$$

$$\sum_{\substack{n \in [1, N] \\ m \in [1, M]}} \left(\sum_{m=1}^M \frac{F_\sigma(k_{n-1}, k_{m-1}) + F_\sigma(k_{n-1}, k_m) + F_\sigma(k, k_{m-1}) + F_\sigma(k_n, k_m)}{4} \right) \quad (\text{A5.14})$$

A5.2 Nondimensionality

$$\int f(x) \frac{1}{\sigma \sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sqrt{2}\sigma}\right)^2} dx \quad (\text{A5.15})$$

$$\chi \rightarrow \frac{x}{\sqrt{2}\sigma} \quad \tilde{\mu} \rightarrow \frac{\mu}{\sqrt{2}\sigma} \quad (\text{A5.16})$$

$$\int f(\sqrt{2}\sigma\chi) e^{-(\chi-\tilde{\mu})^2} d\chi \quad (\text{A5.17})$$

Maybe think in terms of Riemann sums since it is closer to a numerical integral?

$$\sum_{n=\text{TODO}}^N f(x_n) \frac{\Delta x}{\sigma \sqrt{2\pi}} e^{-\left(\frac{x_n-\mu}{\sqrt{2}\sigma}\right)^2} \quad (\text{A5.18})$$

$$\Delta x = \frac{b-a}{N} \quad x_n = x_0 + \Delta x \cdot n \quad (\text{A5.19})$$

If we wish for our x interval to sample over the "interesting" part of the Gaussian and not miss it, we should demand that for every σ the interval need be smaller. If we demand that $\Delta x \leq 0.1\sigma$ we guarantee at least 10 samples inside the significant portion of the integral. More generally, we can set the demand the $\Delta x \leq s \cdot \sigma$ where $0 < s \leq 1$ then

$$s \cdot \sigma \geq \Delta x = \frac{b-a}{N} \implies N \geq \frac{b-a}{s \cdot \sigma} \quad (\text{A5.20})$$

Since the Gaussian is symmetric and vanishes quickly far from μ we can conveniently integrate over an interval of width, say, 10σ or more generally $m \cdot \sigma$ where $m \in \mathbb{N}$ Namely $[\mu - m \cdot \sigma, \mu + m \cdot \sigma]$ which changes the demand on N to be

$$N \geq \left\lceil \frac{2m}{s} \right\rceil \quad (\text{A5.21})$$

The above summation can be then written as

$$\Delta x = \frac{2m\sigma}{2m/s} = s \cdot \sigma \quad \Rightarrow \quad \begin{aligned} x_n &= (\mu - m \cdot \sigma) + (s \cdot \sigma)n \\ &= \mu - \sigma(m - s \cdot n) \end{aligned} \quad (\text{A5.22})$$

This leaves us with the following control parameters:

- σ - gaussian delta approximation
- s - sampling points inside gaussian (essentially Δx)
- m - interval width in terms of σ

A6 Boltzmann Equation

A semi-classical theory treats one part of a system quantum-mechanically and the other classically. In the context of Boltzmann equation, a semi-classical Boltzmann equation treats the particles between collisions classically and the collisions are treated quantum-mechanically.

A6.1 Classical Boltzmann Equation

A PDE whose solution is a single-particle-distribution (SPD). The LHS resembles Liouville equation for an SPD and the RHS has a collision term that accounts for the interaction with the rest of the particles.

$$\frac{\partial f}{\partial t} + \{f, \mathcal{H}_{\text{ext}}\} = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \quad (\text{A6.1})$$

The hamiltonian on the RHS describes the dynamics of the free particle under external influences, and the RHS collision term contains the interaction hamiltonians.

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} f + \mathbf{r} \cdot \nabla_{\mathbf{v}} f = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \quad (\text{A6.2})$$

A6.2 Quantum Boltzmann Equation

$$\frac{\partial f}{\partial t} + \frac{i}{\hbar} [f, \mathcal{H}_{\text{ext}}] = \left(\frac{\partial f}{\partial t}\right)_{\text{coll}} \quad (\text{A6.3})$$

A6.2.1 QBE for Electron Dynamics

$$\left(\frac{\partial f}{\partial t}\right)_{\text{coll}} = \sum_{\text{coll type}} \left(\sum_j f_i f_j W_{i \rightarrow j} \right) \quad (\text{A6.4})$$

Where f_i is the SPD of the first state and f_j is the SPD of the final state summed over all possible final states. $W_{i \rightarrow j}$ is the transition rate given by Fermi golden rule.

A6.3 Relaxation Time Approximation

A6.4 Photon Absorption

$$\cancel{\frac{\partial f}{\partial t}} \overset{\text{steady state}}{\nearrow} = \left(\frac{\partial f}{\partial t}\right)_{\text{ex}} + \left(\frac{\partial f}{\partial t}\right)_{\text{e-e}} + \left(\frac{\partial f}{\partial t}\right)_{\text{e-ph}} \overset{\text{far from } \mathcal{E}_F}{\nearrow} \quad (\text{A6.5})$$

$$\quad (\text{A6.6})$$

$$\Rightarrow \left(\frac{\partial f}{\partial t}\right)_{\text{ex}} = - \left(\frac{\partial f}{\partial t}\right)_{\text{e-e}} \quad (\text{A6.7})$$

Technically, there should also be a term for the recombination (and emission) of electrons, but it is very negligible in metals (according to Yonatan - need to check why). An involved calculation leads to the excitation (absorption) term

$$\left(\frac{\partial f}{\partial t}\right)_{\text{ex}} = R(\omega_L)|E_L|^2 \cdot B(\mathcal{E}; \omega_L) \quad (\text{A6.8})$$

with E_L as the electric field strength of the pump and R is a function of the pump photon's frequency (see explicit expressions for R and B below). Relaxation time approximation gives

$$\left(\frac{\partial f}{\partial t}\right)_{\text{e-e}} \approx \frac{f^T - f}{\tau_{e-e}} \equiv -\frac{\Delta f}{\tau_{e-e}} \quad (\text{A6.9})$$

Plugging Eq. (A6.9) and Eq. (A6.8) into Eq. (A6.7) results in

$$\Delta f = \tau_{e-e} R(\omega_L) |E_L|^2 \cdot B(\mathcal{E}, \omega_L) \quad (\text{A6.10})$$

$$\quad (\text{A6.11})$$

$$\equiv \delta_E \cdot B(\mathcal{E}; \omega_L) \quad (\text{A6.12})$$

(see the definition of δ_E below) Finally, the non-equilibrium distribution is

$$f(\mathcal{E}; \omega_L) = f^T(\mathcal{E}) + \underbrace{\delta E(\omega_L) \cdot B(\mathcal{E}, \omega_L)}_{f^{NT}(\mathcal{E}; \omega_L)} \quad (\text{A6.13})$$

Note: Definitions of B , δ_E and R

$$B(\mathcal{E}; \omega_L) = f(\mathcal{E} - \hbar\omega_L) [1 - f(\mathcal{E})] - f(\mathcal{E}) [1 - f(\mathcal{E} + \hbar\omega_L)] \quad (\text{A6.14})$$

$$\quad (\text{A6.15})$$

$$\delta E \equiv \left| \frac{E_L}{E_{\text{sat}}} \right|^2 = |\tau_{e-e} R(\omega_L) E_L|^2 \quad (\text{A6.16})$$

$$\quad (\text{A6.17})$$

$$R(\omega_L) = \frac{4\epsilon_0 \epsilon_m''(\omega_L)}{3\hbar n_e} \frac{\mathcal{E}_F}{\hbar\omega_L} \quad (\text{A6.18})$$

Need to read about the source of these expressions

A7 Quadratic Intraband Transitions

A7.1 Conduction Band

The conduction band as given by 5 - Rosei is as follows

$$\mathcal{E}_c(\mathbf{k}) = \mathcal{E}_{0c} + \frac{\hbar^2 k_\perp^2}{2m_{c\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{c\parallel}} \quad (\text{A7.1})$$

For intraband transitions it is natural too set $\mathcal{E}_{0c} = 0$, and reassign the quadratic coefficients which leaves us with the following dispersion relation

$$\mathcal{E}_c(\mathbf{k}) = Ak_\perp^2 - Bk_\parallel^2 \quad (\text{A7.2})$$

A7.1.1 General Derivation

$$I(\hbar\omega) \propto \iint_{\text{BZ}} d^3k_1 d^3k_2 f(\mathcal{E}_c(\mathbf{k}_1)) [1 - f(\mathcal{E}_c(\mathbf{k}_2))] \delta(\mathcal{E}_c(\mathbf{k}_1) - \mathcal{E}_c(\mathbf{k}_2) - \hbar\omega) \quad (\text{A7.3})$$

Unlike the inter-band case, this process is phonon-assisted (or defect-assisted), meaning momentum is **not conserved** ($\mathbf{k}_1 \neq \mathbf{k}_2$) even approximately. We assume that whatever momentum is missing for an energetic transition to occur is readily available through various mediators (as mentioned above). The initial and final states are independent vectors in the Brillouin Zone. This allows us to decouple the six-dimensional integral into a product of two independent densities of states. By using the following

$$\delta(\mathcal{E}_c(\mathbf{k}_1) - \mathcal{E}_c(\mathbf{k}_2) - \hbar\omega) = \iint \delta(\mathcal{E}_1 - \mathcal{E}_c(\mathbf{k}_1)) \times \delta(\mathcal{E}_2 - \mathcal{E}_c(\mathbf{k}_2)) \times \delta(\mathcal{E}_1 - \mathcal{E}_2 - \hbar\omega) d\mathcal{E}_1 d\mathcal{E}_2 \quad (\text{A7.4})$$

We can write

$$I(\hbar\omega) \propto \int_{-\infty}^{\infty} d\mathcal{E}_1 d\mathcal{E}_2 f(\mathcal{E}_1) [1 - f(\mathcal{E}_2)] \delta(\mathcal{E}_1 - \mathcal{E}_2 - \hbar\omega) \quad (\text{A7.5})$$

$$\times \underbrace{\left(\int_{\text{BZ}} d^3k_1 \delta(\mathcal{E}_1 - \mathcal{E}_c(\mathbf{k}_1)) \right)}_{\rho_c(\mathcal{E}_1)} \underbrace{\left(\int_{\text{BZ}} d^3k_2 \delta(\mathcal{E}_2 - \mathcal{E}_c(\mathbf{k}_2)) \right)}_{\rho_c(\mathcal{E}_2)} \quad (\text{A7.6})$$

and we retrieve the familiar 1D expression for emission

$$I(\hbar\omega) \propto \int_{-\infty}^{\infty} f(\mathcal{E}) [1 - f(\mathcal{E} - \hbar\omega)] \rho_c(\mathcal{E}) \rho_c(\mathcal{E} - \hbar\omega) d\mathcal{E} \quad (\text{A7.7})$$

The challenge here is to properly find a closed expression for the electronic density of states (eDOS).

A7.2 Finding the Conduction Band eDOS

First COV (Cylindrical Symmetry) - Using the rotational symmetry of the saddle point near X

$$d^3k \rightarrow 2\pi k_\perp dk_\perp dk_\parallel \quad (\text{A7.8})$$

Second COV (Linearization) - We introduce variables u and v to linearize the energy expression

$$u = Ak_{\perp}^2 \implies \frac{du}{A} = 2k_{\perp} dk_{\perp} \quad (\text{A7.9})$$

$$v = Bk_{\parallel}^2 \implies \frac{1}{2\sqrt{B}} \frac{dv}{\sqrt{v}} = dk_{\parallel} \quad (\text{A7.10})$$

Accounting for the symmetry of $k_{\parallel}$ (integrating from $-\infty$ to ∞ is equivalent to $2 \times$ integrating 0 to ∞), the volume element becomes

$$d^3k \rightarrow \frac{\pi}{A\sqrt{B}} \frac{dudv}{\sqrt{v}} \quad (\text{A7.11})$$

Next, we find the proper integration limits on u

$$0 \leq k_{\parallel} \leq k_{max} \implies 0 \leq v \leq \underbrace{Bk_{max}^2}_{\mathcal{E}_{\parallel}^{\Phi \rightarrow \S}} \quad (\text{A7.12})$$

$$0 \leq k_{\perp} \leq k_{max} \implies \underbrace{\max[0, \mathcal{E}_j]}_{\mathcal{E}_{\perp}^{\Phi} \setminus (\mathcal{E}_{\parallel})} \leq u \leq \underbrace{Ak_{max}^2}_{\mathcal{E}_{\perp}^{\Phi \rightarrow \S}} \quad (\text{A7.13})$$

Note: (b) $\rightarrow$ (c) transition

The delta function forces its argument to vanish

$$\mathcal{E}_j - \mathcal{E}_c(u, v) \stackrel{!}{=} 0 \quad (\text{A7.14})$$

$$\mathcal{E}_j - u + v \stackrel{!}{=} 0 \quad (\text{A7.15})$$

$$v - (u - \mathcal{E}_j) \stackrel{!}{=} 0 \quad (\text{A7.16})$$

The subscript $j = 1, 2$ refers to the initial and final state energies. All three equations serve as equivalent arguments for the delta function. To eliminate the v component we chose the last.

Let us apply the COV to the density of states

$$\rho(\mathcal{E}_j) = \int_{\text{BZ}} \delta(\mathcal{E}_j - \mathcal{E}_c(\mathbf{k})) d^3k \quad (\text{a}) \quad (\text{A7.17})$$

$$= \frac{\pi}{A\sqrt{B}} \int_{\mathcal{E}_{\perp}^{\Phi} \setminus (\mathcal{E}_{\parallel})}^{\mathcal{E}_{\perp}^{max}} \int_0^{\mathcal{E}_{\parallel}^{\Phi \rightarrow \S}} \delta[\mathcal{E}_j - \mathcal{E}_c(u, v)] \frac{dudv}{\sqrt{v}} \quad (\text{b}) \quad (\text{A7.18})$$

$$= \frac{\pi}{A\sqrt{B}} \int_{\mathcal{E}_{\perp}^{\Phi} \setminus (\mathcal{E}_{\parallel})}^{\mathcal{E}_{\perp}^{max}} \int_0^{\mathcal{E}_{\parallel}^{\Phi \rightarrow \S}} \delta[v - (u - \mathcal{E}_j)] \frac{dudv}{\sqrt{v}} \quad (\text{c}) \quad (\text{A7.19})$$

$$= \frac{\pi}{A\sqrt{B}} \int_{\mathcal{E}_{\perp}^{\Phi} \setminus (\mathcal{E}_{\parallel})}^{\mathcal{E}_{\perp}^{max}} \frac{du}{\sqrt{u - \mathcal{E}_j}} \quad (\text{d}) \quad (\text{A7.20})$$

Note: Integration Limits

First, lets us express v in terms of u

$$v_j(u) = u - \mathcal{E}_j \quad (\text{A7.21})$$

we apply both integration limits brought from $k_\perp$ and $k_\parallel$

$$k_\perp^2 \geq 0 \implies u \geq 0 \quad (\text{A7.22})$$

$$k_\parallel^2 \geq 0 \implies v(u) \geq 0 \quad (\text{A7.23})$$

we apply both conditions to u

$$u \geq \mathcal{E}_j \quad u \geq 0 \quad (\text{A7.24})$$

This provides a lower limit on the u integration, and we choose the the upper lower bound, namely

$$u_{min}(\mathcal{E}_j) = \max[0, \mathcal{E}_j] \quad (\text{A7.25})$$

$$\rho(\mathcal{E}_j) = \frac{\pi}{A\sqrt{B}} \int_{\mathcal{E}_\perp^{\Phi} \setminus (\mathcal{E}_\parallel)}^{\mathcal{E}_\perp^{max}} \frac{du}{\sqrt{u - \mathcal{E}_j}} \quad (\text{A7.26})$$

$$= \frac{2\pi}{A\sqrt{B}} \left[\sqrt{\mathcal{E}_\perp^{max} - \mathcal{E}_\parallel} - \sqrt{\mathcal{E}_\perp^{\Phi} \setminus (\mathcal{E}_\parallel) - \mathcal{E}_j} \right] \quad (\text{A7.27})$$

$$= \frac{2\pi}{A\sqrt{B}} \left[\sqrt{\mathcal{E}_\perp^{max} - \mathcal{E}_\parallel} - \sqrt{\max(0, \mathcal{E}_j) - \mathcal{E}_j} \right] \quad (\text{A7.28})$$

A7.2.1 Putting it All Together

$$I(\hbar\omega) \propto \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d\mathcal{E}_1 d\mathcal{E}_2 f(\mathcal{E}_1) [1 - f(\mathcal{E}_2)] \delta(\mathcal{E}_1 - \mathcal{E}_2 - \hbar\omega) \times \quad (\text{A7.29})$$

$$\times \frac{2\pi}{A\sqrt{B}} \left[\sqrt{\mathcal{E}_\perp^{max} - \mathcal{E}_1} - \sqrt{\max[0, \mathcal{E}_1] - \mathcal{E}_1} \right] \quad (\text{A7.30})$$

$$\times \frac{2\pi}{A\sqrt{B}} \left[\sqrt{\mathcal{E}_\perp^{max} - \mathcal{E}_2} - \sqrt{\max[0, \mathcal{E}_2] - \mathcal{E}_2} \right] \quad (\text{A7.31})$$

After resolving the delta function for $\mathcal{E}_2$, and setting $\mathcal{E}_1 \equiv \mathcal{E}$, we're left with

$$\begin{aligned}
 I(\hbar\omega) \propto & \left(\frac{2\pi}{A\sqrt{B}} \right)^2 \int_{-\infty}^{\infty} d\mathcal{E} f(\mathcal{E}) \left[1 - f(\mathcal{E} - \hbar\omega) \right] \times \\
 & \times \left[\sqrt{\mathcal{E}_{\perp}^{max} - \mathcal{E}} - \sqrt{\max[0, \mathcal{E}] - \mathcal{E}} \right] \\
 & \times \left[\sqrt{\mathcal{E}_{\perp}^{max} - \mathcal{E} + \hbar\omega} - \sqrt{\max[0, \mathcal{E} - \hbar\omega] - \mathcal{E} + \hbar\omega} \right]
 \end{aligned} \tag{A7.32}$$

A8 Interband Transitions

We begin by adopting the band approximation Given by Rossei for the valence and conduction dispersion relations,

$$\mathcal{E}_v(\mathbf{k}) = -\mathcal{E}_{0v} - \frac{\hbar^2 k_\perp^2}{2m_{V\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{V\parallel}} \implies -\mathcal{E}_{0v} - A_v k_\perp^2 - B_v k_\parallel^2 \quad (\mathcal{E} < \mathcal{E}_{0v}) \quad (\text{A8.1})$$

$$\mathcal{E}_c(\mathbf{k}) = \mathcal{E}_{0c} + \frac{\hbar^2 k_\perp^2}{2m_{C\perp}} - \frac{\hbar^2 k_\parallel^2}{2m_{C\parallel}} \implies \mathcal{E}_{0c} + A_c k_\perp^2 - B_c k_\parallel^2 \quad (\mathcal{E} > \mathcal{E}_{0c}) \quad (\text{A8.2})$$

Our starting point is the interband emission integral given by Fermi's golden rule for the transition from an initial momentum state $\mathbf{k}_1$ to a final state $\mathbf{k}_2$

$$\Gamma_e^{\text{cv}}(\hbar\omega) \propto \int_{\text{BZ}} d^3 k_1 \int_{\text{BZ}} d^3 k_2 f(\mathcal{E}_c(\mathbf{k}_1)) [1 - f(\mathcal{E}_v(\mathbf{k}_2))] \delta(\mathcal{E}_c(\mathbf{k}_1) - \mathcal{E}_v(\mathbf{k}_2) - \hbar\omega) \quad (\text{A8.3})$$

We consider only direct interband transitions $\mathbf{k}_1 \approx \mathbf{k}_2$. The integral loses three degrees of freedom and the transition effectively takes place at a single $\mathbf{k}$ in phase space, giving

$$\Gamma_e^{\text{cv}}(\hbar\omega) \propto \int_{\text{BZ}} d^3 k f(\mathcal{E}_c(\mathbf{k})) [1 - f(\mathcal{E}_v(\mathbf{k}))] \delta(\mathcal{E}_c(\mathbf{k}) - \mathcal{E}_v(\mathbf{k}) - \hbar\omega) \quad (\text{A8.4})$$

The rest of the derivation is a sequence of changes of variables that makes the delta function explicit and reduces the integral to one dimension.

A8.1 1. Cylindrical Symmetry

The first change of variables is a trivial one that adopts the cylindrical symmetry in k-space near the X and L points.

$$dk_x dk_y dk_z \rightarrow k_\perp dk_\perp dk_\parallel dk_\phi \rightarrow 2\pi k_\perp dk_\perp dk_\parallel \quad (\text{A8.5})$$

A8.2 2. Linearized Momenta

Next, we linearize the dispersion relation by performing yet another change of variables

$$u = k_\perp^2 \implies du = 2k_\perp dk_\perp \quad k_\perp \in [0, \infty) \rightarrow u \in [0, \infty) \quad (\text{A8.6})$$

$$v = k_\parallel^2 \implies \frac{1}{2} \frac{dv}{\pm\sqrt{v}} = dk_\parallel \quad k_\parallel \in (-\infty, \infty) \rightarrow v \in [0, \infty) \quad (\text{A8.7})$$

To account for $\pm\sqrt{v}$ we can use the fact that the bands are even functions in momentum, and are therefore symmetric with respect to $k_\parallel$. The Jacobean is multiplied by a factor of 2 to account for the two contributions

$$2\pi k_\perp dk_\perp dk_\parallel \rightarrow \pi \frac{du dv}{\sqrt{v}} \quad (\text{A8.8})$$

A8.3 Energy Space

Finally, we introduce another linear transformation from the linearized momenta u and v into the more physically meaningful variables $\mathcal{E}$ and Δ which are the conduction band energy and the energy difference between the conduction and valence

$$\mathcal{E} \equiv \mathcal{E}_c(u, v) \quad \Longrightarrow \quad \mathcal{E} - \mathcal{E}_{c0} = A_c u - B_c v \quad (\text{A8.9})$$

$$\Delta \equiv \mathcal{E}_c(u, v) - \mathcal{E}_v(u, v) \quad \Longrightarrow \quad \Delta - \mathcal{E}_g = \bar{A}u + \bar{B}v \quad (\text{A8.10})$$

where, $\mathcal{E}_g \equiv \mathcal{E}_{c0} + \mathcal{E}_{v0}$ $\bar{A} \equiv A_c + A_v$ $\bar{B} \equiv B_v - B_c$. To find the Jacobean, the integration limits in terms of the new variables and the inverse transformation we write the linear change of variables in matrix notation

$$\begin{pmatrix} \mathcal{E} - \mathcal{E}_{c0} \\ \Delta - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_c & -B_c \\ \bar{A} & \bar{B} \end{pmatrix}}_M \begin{pmatrix} u \\ v \end{pmatrix} \quad (\text{A8.11})$$

The inverse transformation is given by

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{D} \begin{pmatrix} \bar{B} & B_c \\ -\bar{A} & A_c \end{pmatrix} \begin{pmatrix} \mathcal{E} - \mathcal{E}_{c0} \\ \Delta - \mathcal{E}_g \end{pmatrix} \quad (\text{A8.12})$$

This allows us to extract $u(\mathcal{E}, \Delta)$ and $v(\mathcal{E}, \Delta)$.

$$u(\mathcal{E}, \Delta) = \frac{1}{D} [\bar{B}(\mathcal{E} - \mathcal{E}_{c0}) + B_c(\Delta - \mathcal{E}_g)] \quad (\text{A8.13})$$

$$v(\mathcal{E}, \Delta) = \frac{1}{D} [-\bar{A}(\mathcal{E} - \mathcal{E}_{c0}) + A_c(\Delta - \mathcal{E}_g)] \quad (\text{A8.14})$$

Integration limits can be found by posing the constraints $u > 0$ and $v > 0$ deduced in the previous transformation

$$\mathcal{E}_{max}(\Delta) = \mathcal{E}_{c0} + \frac{A_c}{A_c + A_v}(\Delta - \mathcal{E}_g) \quad (\text{A8.15})$$

$$\mathcal{E}_{min}(\Delta) = \mathcal{E}_{c0} - \frac{B_c}{B_v + B_c}(\Delta - \mathcal{E}_g) \quad (\text{A8.16})$$

We can now neatly write $v(\mathcal{E}, \Delta)$ in terms of $\mathcal{E}_{max}$

$$v(\mathcal{E}, \Delta) = \frac{\bar{A}}{D} (\mathcal{E}_{max}(\Delta) - \mathcal{E}) \quad (\text{A8.17})$$

and insert it into the transformation (including the $1/D$ Jacobean factor from this current transformation). The new measure becomes

$$\pi \frac{dudv}{\sqrt{v}} \rightarrow \frac{\pi}{\sqrt{AD}} \frac{d\Delta d\mathcal{E}}{\sqrt{\mathcal{E}_{max}(\Delta) - \mathcal{E}}} \quad (\text{A8.18})$$

Putting everything together, the emission integral turns into

$$\Gamma_e^{cv}(\hbar\omega) \propto \int_{\text{BZ}} f(\overbrace{\mathcal{E}_c(\mathbf{k})}^{\mathcal{E}}) \left[1 - f(\overbrace{\mathcal{E}_v(\mathbf{k})}^{\mathcal{E}-\cdot}) \right] \delta\left(\overbrace{\mathcal{E}_c(\mathbf{k}) - \mathcal{E}_v(\mathbf{k})}^{\Delta} - \hbar\omega\right) \underbrace{\frac{d^3k}{\sqrt{DA} \sqrt{\mathcal{E}_{max}(\Delta) - \mathcal{E}}}}_{\text{A8.19}} \quad (\text{A8.19})$$

The delta function, now explicit in its arguments with respect to the integration variables can finally be resolved, eliminating the Δ integral, leaving a one-dimensional expression

$$\Gamma_e^{\text{cv}}(\hbar\omega) = \frac{\pi^2 |\mu|^2}{\hbar \sqrt{AD}} \int_{\mathcal{E}_{\min}(\hbar\omega)}^{\mathcal{E}_{\max}(\hbar\omega)} \frac{f(\mathcal{E}) f(\hbar\omega - \mathcal{E})}{\sqrt{\mathcal{E}_{\max}(\hbar\omega) - \mathcal{E}}} d\mathcal{E} \quad (\text{A8.20})$$

which still must be solved numerically, but is a one-dimensional rather than a three-dimensional integral and it avoids the problematic delta function.

A9 Kirchhoff's Law Of Radiation

In thermodynamics equilibrium (TDE) a material body emits according to the expression

$$\mathcal{E}(\omega) = A(\omega) \cdot u_{\text{BB}}(\omega) \quad (\text{A9.1})$$

The Planck Blackbody law is composed of two distinct parts:

$$u_{\text{BB}}(\omega) = \underbrace{\frac{\hbar\omega^3}{\pi^2 c^3}}_{\text{Photonic}} \times \underbrace{\frac{1}{e^{\hbar\omega/k_B T} - 1}}_{\text{Statistical}} \quad (\text{A9.2})$$

The Photonic Part (ω^3) Comes from the ratio of vacuum modes to material absorption cross-section **The Statistical Part** ($e^x - 1$) Comes exactly from the ratio of the Fermi factors.

Here is the mathematical proof that your Fermi factors generate the Bose-Einstein statistics necessary for Kirchhoff's Law.

A9.1 From Fermi-Dirac to Bose-Einstein

- **Spontaneous Emission Rate** (Γ_{sp}): Electron starts at $\mathcal{E} + \hbar\omega$ (Upper) and ends at $\mathcal{E}$ (Lower).

$$\Gamma_{sp} \propto f(\mathcal{E} + \hbar\omega)[1 - f(\mathcal{E})] \quad (\text{A9.3})$$

- **Net Absorption Rate** (Γ_{abs}): This is tricky. To get the exact Planck law, you must calculate **Net Absorption** (Absorption minus Stimulated Emission).

$$\Gamma_{abs} \propto \underbrace{f(\mathcal{E})[1 - f(\mathcal{E} + \hbar\omega)]}_{\text{Absorption}} - \underbrace{f(\mathcal{E} + \hbar\omega)[1 - f(\mathcal{E})]}_{\text{Stimulated Emission}} \quad (\text{A9.4})$$

Now, let's take the ratio of **Emission to Net Absorption** and see if we get the Planck statistical factor.

$$\text{Ratio} = \frac{\text{Spontaneous}}{\text{Absorption} - \text{Stimulated}} \quad (\text{A9.5})$$

$$\text{Ratio} = \frac{f(U)[1 - f(L)]}{f(L)[1 - f(U)] - f(U)[1 - f(L)]} \quad (\text{A9.6})$$

(Using shorthand: $U = \mathcal{E} + \hbar\omega$, $L = \mathcal{E}$)

Divide the numerator and denominator by the numerator ($f(U)[1 - f(L)]$):

$$\text{Ratio} = \frac{1}{\frac{f(L)[1 - f(U)]}{f(U)[1 - f(L)]} - 1} \quad (\text{A9.7})$$

A9.2 The Magic of the Fermi Function

Now we evaluate the fraction in the denominator using the definition $f(\mathcal{E}) = \frac{1}{e^{(\mathcal{E} - \mathcal{E}_F)/k_B T} + 1}$.

A useful identity for Fermi functions is:

$$\frac{1 - f(\mathcal{E})}{f(\mathcal{E})} = e^{(\mathcal{E} - \mathcal{E}_F)/k_B T} \quad (\text{A9.8})$$

Let's plug this into our fraction:

$$\frac{f(L)[1 - f(U)]}{f(U)[1 - f(L)]} = \left(\frac{f(L)}{1 - f(L)} \right) \times \left(\frac{1 - f(U)}{f(U)} \right) \quad (\text{A9.9})$$

$$= \left(e^{-(L - \mathcal{E}_F)/kT} \right) \times \left(e^{(U - \mathcal{E}_F)/kT} \right) \quad (\text{A9.10})$$

$$= e^{(U - L)/k_B T} \quad (\text{A9.11})$$

Since $U - L = (\mathcal{E} + \hbar\omega) - \mathcal{E} = \hbar\omega$, the fraction becomes exactly:

$$e^{\hbar\omega/k_B T} \quad (\text{A9.12})$$

A9.3 The Result

Substitute this back into our Ratio equation:

$$\text{Ratio} = \frac{1}{e^{\hbar\omega/k_B T} - 1} \quad (\text{A9.13})$$

This is exactly the **Bose-Einstein distribution**.

A9.4 Conclusion

Your intuition regarding the Fermi factors is correct and crucial.

- **The ω^3 factor** (from the Density of States) explains why the Blackbody spectrum rises polynomially at low frequencies.
- **The Fermi factor ratio** (which you identified) explains why it rolls over and dies exponentially at high frequencies ($e^{-\hbar\omega/kT}$).

Both are required to perfectly reproduce the Planck law from microscopic principles.

A9.5 The Role of Stimulated Emission

It is worth noting that this derivation fails to produce the Planck law if **stimulated emission** is neglected. If we were to compare spontaneous emission directly to absorption (ignoring the stimulated term in the denominator), the ratio would simplify to:

$$\text{Ratio} = \frac{\Gamma_{sp}}{\Gamma_{abs, total}} = \frac{f(U)[1 - f(L)]}{f(L)[1 - f(U)]} = e^{-\hbar\omega/k_B T} \quad (\text{A9.14})$$

This result is known as **Wien's approximation**. While it correctly describes the high-energy (exponential decay) tail of the spectrum, it fails at low energies because it misses the "−1" in the denominator.

Physically, stimulated emission is the mechanism that accounts for the **bosonic nature** of photons. Neglecting it is equivalent to treating photons as classical, distinguishable particles (Maxwell-Boltzmann statistics) rather than Bosons. To recover the full **Bose-Einstein distribution**, the net absorption must account for the photons' tendency to induce further transitions.

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